

the limitations of this particular design, home assembly was abandoned in favor of higher performance necessary to run an operating system.

In the next couple of years the boom in smartphones indicated that an affordable processor unit could form the backbone of the Raspberry Pi design. As custom microprocessors were being developed at Upton's own employer - Broadcom - affordability and performance were both becoming realistic for the Pi. The next iteration of the device abandoned all homemade elements like veroboards, DIP chips and through-hole components and looked to fine crafted manufacturing assembly. The first alpha design board was sent for manufacturing in July 2011, just months before its February 2012 launch. The board was identical to what would be the final release with exceptions in its slightly larger size and more layers. It used an ARM processor along with an Ethernet controller. These alpha boards were used primarily to run software applications testing. Within a month of the alpha board testing the Raspberry Pi team was successfully running Quake 3 Deathmatch on a set of networked devices in the office.

The next major development was with refining the power distribution layout of the board's design. The challenge remained to keep the power consumption stable while ensuring a stable route for all components. This problem could have been easily solved if the team was willing to spend more money on the board designs. But given their commitment to keeping the price point low they attempted to figure out another solutions. Eventually, the team arrived at the solution of using microscopic laser made holes in the circuit board to evenly distribute board space for different electrical components. The wiring channels or micro vias would only increase the per unit price by pennies when done at a large scale using lasers. The ability to wire the printed circuit boards or PCBs at different levels of its thickness allowed the final design to become fully functional.

The basic layout and design of the Raspberry Pi remained fairly unchanged for the bulk of its development. Once the team had discovered the right low cost ARM processors for their needs, the rest of the board layout and design was easy to execute. The real challenge was towards mass production and business partnerships which would allow them to leverage economies of scale and allow their price points to remain unchanged. The final beta test boards for the Model B were wrapped up in December 2011 and ten of them were auctioned off online. The final production line for the Model B started off in January 2012, while the first root filesystem disc image was developed for Debian release in February. The preorders for the Rasp-